



THE UNIVERSITY OF
SYDNEY

—
Core Research
Facilities

—
Sydney
Informatics Hub

The Sydney Informatics Hub presents

Accelerated Computing for Innovation Conference

Friday 28 September 2018

Join us to celebrate innovation driven by cutting-edge tech, big data, machine learning and AI. Listen to fascinating talks from science and industry about data science and technology powering precision medicine, wildlife and crop conservation, virtual earth explorations, rock and shark spotting, education and language tech, crime prediction, and more! Leave full of ideas, connected, and motivated.

Free registration

informatics.sydney.edu.au/hpc_conference/

Location

ABS Lecture Theatre 1130
Abercrombie Building (H70)
cnr Abercrombie St and Codrington St
University of Sydney

Program

8.45am - 9.00am Registration
9.00am - 9.10am Welcome
9.10am - 11.15am Life, Evolution and Health
11.15am - 11.30am Morning tea
11.30am - 12.30pm National Collaborative Infrastructure
12.30pm - 1.15pm Lunch
1.15pm - 3.15pm Earth, Environment and Space
3.15pm - 3.30pm Afternoon tea
3.30pm - 5.35pm Humans and Society
5.35pm - 5.45pm Closing Remarks

More information

Sydney Informatics Hub

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Office of the
Chief Scientist
& Engineer





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Program

Time	Session
8.45am – 9.00am	Registration
9.00am – 9.10am	Opening address
9.00am – 9.10am	Jeremy Hammond , USyd
9.10am – 11.15am	Life, Evolution and Health
9.10am – 9.35am	Aiden O'Brien , CSIRO - Life science research at scale
9.35am – 10.00am	Mark Pinese , Garvan - Computing the genomics of good health
10.00am – 10.25am	Carolyn Hogg , USyd - The future is here - the benefits of genomic tools in translocations
10.25am – 10.50am	John Sebastian Eden , WIMR, USyd - Revealing the hidden world of viruses
10.50am – 11.15am	Emily Remnant , USyd - Using big data to find small things inside bees
11.15am – 11.30am	Morning tea (provided)
11.30am – 1.15pm	National Collaborative Infrastructure
11.30am – 11.50am	Andrew Lonie , UniMelb - Planning digital research infrastructure for biosciences
11.50am – 12.10pm	Christopher Harris , Pawsey - Enabling Petascale Science with the Pawsey Supercomputing Centre
12.10pm – 12.30pm	Ben Evans , NCI - HPC and data science at NCI - enabling peak science and collaboration
12.30pm – 1.15pm	Lunch (provided)

Time	Session
1.15pm – 3.15pm	Earth, Environment and Space
1.15pm – 1.45pm	Dietmar Muller , USyd - Understanding Earth System Evolution - connecting Deep to Surface Processes
1.45pm – 2.15pm	Geraint Lewis , USyd - What's happening in the dark-side of the universe?
2.15pm – 2.45pm	Caroline Foster , USyd - Galaxy formation and evolution
2.45pm – 3.15pm	Cormac Purcell , MQ - Sharks, tombs and cosmic bubbles: how machine-learning spans diverse scientific fields
3.15pm – 3.30pm	Afternoon tea (provided)
3.30pm – 5.30pm	Humans and Society
3.30pm – 3.55pm	Kai Riemer and Sandra Peter , USyd - AI - History, hype and hysteria
3.55pm – 4.20pm	Simon Hammond , Appen - Understanding speech and text in low-resource languages
4.20pm – 4.45pm	Bing Ong , Daisee
4.45pm – 5.10pm	Sonya Corcoran , Digital Innovation - Who is making what and how?
5.10pm – 5.35pm	Roman Marchant , USyd - Understanding and preventing crime using Data Science
5.35pm – 5.45pm	Closing Remarks
5.35pm – 5.45pm	Simon Ringer , USyd

Our **speakers** are supported by:



The **Accelerated Computing for Innovation Conference** is sponsored and supported by:



Life, Evolution and Health 9:15-11:15am

Aiden O'Brien 9:10am - 9:35am



Life science research at scale

Genomic data is outpacing traditional big data disciplines, producing more information than astronomy, Twitter, and YouTube combined. As such, genomic research has leapfrogged to the forefront of big data, requiring novel cloud solutions with artificial intelligence and machine learning to generate useful insights. This talk therefore showcases how we find disease genes responsible for ALS using VariantSpark, a random forest implementation built on Spark to deal with the 80 million dimensions in genomic data. Next, this talk outlines how we use a serverless architecture to translate these insights into the clinical practice by providing a decision support framework for clinicians. Finally, the talk touches on how to evolve serverless architecture more efficiently through a hypothesis-driven approach to DevOps and how we keep data and functions secure in a serverless environment.

Aidan O'Brien is a PhD student with the **Transformational Bioinformatics team, CSIRO** and the Australian National University. He develops machine learning models for accurate genome editing as needed in gene therapy for human health. He developed GT-Scan, a CRISPR target predictor and is a developer on VariantSpark, which enables personalized genomic insights at point-of-care by finding 'patients with conditions like mine' based on their genomic similarity. He has an enthusiasm for serverless (FaaS) and hypothesis-driven architecture design, which he presented at YOW! Data 2018. He received the "Best student and postdoc" award at CSIRO in 2015.

Mark Pinese 10:25am - 10:50am

Computing the genomics of good health

Abstract

Dr Mark Pinese is a Senior Research Officer in the **Garvan's Genomic Cancer Medicine Lab** where he is the lead analyst for the Medical Genome Reference Bank and conducts research into the genetic basis of cancer risk.



Carolyn Hogg 10am - 10:25am

The future is here - the benefits of genomic tools in translocations

Abstract

Dr Carolyn Hogg is the Research Manager for the **Australasian Wildlife Genomics Group** in the Faculty of Science, University of Sydney. Carolyn is a conservation biologist who has been working with threatened species for over twenty years both in Australia and overseas. Carolyn has been working closely with the Save the Tasmanian Devil Program for over eight years and with other threatened species more recently. Through her partnerships with both her academic and conservation colleagues Carolyn has developed better tools and technologies to integrate molecular genetics into real-time conservation management decisions.



John Sebastian Eden 10:25am – 10:50am



**Revealing the hidden world of viruses,
one Giga-base pair at a time.**

The field of pathogen genomics has been transformed by new sequencing technologies. From practically any sample, billions of sequence reads can be generated that provide an unbiased snapshot of what viruses, bacteria, fungi, and other parasites, might be present. Meta-transcriptomics or bulk RNA sequencing has proven to be particularly powerful for viral discovery and our research highlights the richness of this data. However, the ease at which data can be produced is contrasted sharply against the computational and analytical challenges in handling the volumes of data as well as being able to accurately describe genomes which are incredibly diverse. Here, I will present data from our recent metagenomic studies and outline how we utilise high-performing computing to address some of the previously mentioned challenges.

Dr John-Sebastian Eden is a Research Scientist in Bioinformatics and Genomics at the **Westmead Institute for Medical Research** and Senior Research Fellow in the **Sydney Medical School**. Previously, he was an NHMRC Early Career Fellow working with Prof Eddie Holmes at the University of Sydney. His research uses pathogen genomics and phylodynamics to better understand the sources and mechanisms of how pathogens emerge, spread and evolve in populations. Recent work includes the development of novel assays for sequencing individual HIV proviruses to explore the latent reservoir and the application of “meta-transcriptomics” to pathogen discovery in both human and wildlife disease investigations.

Emily Remnant 10:50am - 11:15am

Using big data to find small things inside bees

Abstract

Dr Emily Remnant is a geneticist and evolutionary biologist in the **Social Insects Lab, University of Sydney** investigating how insects respond to changes in their environment. She use genetics and genomics to study genes involved in resistance to chemical insecticides, and to investigate the impact of viral diseases in beneficial insects like honey bees on crop pollination.



National Collaborative Infrastructure 11:30am-1:15pm

Andrew Lonie 11:30am - 10:50am

Planning digital research infrastructure for biosciences



Some research disciplines have advanced, multi-year community-sourced roadmaps for digital infrastructure in Australia; unfortunately, biosciences isn't one of those disciplines.

Bioplatforms Australia, through the EMBL Australia Bioinformatics Resource and associates, is working to define a national 'bioinformatics commons' – national research infrastructure to support widespread access to the critical data, tools, compute, techniques and training required for high quality analysis of digital data in biosciences. This means understanding the different types of researchers using and doing bioinformatics, what their needs are, what is happening globally, and how we can work together as a country to benefit locally and engage internationally. If done properly, we believe this work will inform and influence national infrastructure investments in this space long term, particularly the Australian Research Data Commons (<https://www.ands-nectar-rds.org.au/>).

In this talk I will provide an overview of the bioinformatics commons project and discuss the process and progress to date, including engaging with the bioinformatics community and future plans.

Andrew Lonie is Director of **Melbourne Bioinformatics** and an associate professor at the Faculty of Medicine, Dentistry and Health Sciences at the University of Melbourne, where he coordinates the MSc (Bioinformatics). Andrew directs a group of bioinformaticians, computational biologists and HPC specialists to collaborate with and support life sciences researchers in a variety of research projects within the Parkville research precinct and across Australia.

Christopher Harris 11.50am – 12.10pm

**Enabling Petascale Science with the
Pawsey Supercomputing Centre**



The Pawsey Supercomputing Centre encourages and energises research using supercomputing, large scale data storage and visualisation. We provide facilities and expertise to research, education and industrial communities across Australia. Application areas include nanotechnology, radio astronomy, high energy physics, medical research, mining and petroleum, architecture and construction, multimedia, and urban planning, and many more.

This presentation will provide an overview of Pawsey facilities that are available nationally to Australian researchers, including details of how to get access, as well as showcase some of the currently supported research projects.

Dr Christopher Harris is a Senior Supercomputing Specialist at the Pawsey Supercomputing Centre. He provides assistance and expertise to researchers, enabling computational workflows to make effective use of Australia's national petascale HPC facilities.

Ben Evans 12:10 – 12:30pm

**HPC and data science at NCI - enabling peak science and
collaboration**

Abstract

Earth, Environment and Space 1:15-3:15pm

Dietmar Müller 1:15pm - 1:45pm

Understanding Earth System Evolution -connecting Deep to Surface Processes



Our planet is of a composition and a location in space that has resulted in a thermal and structural evolution that is unique in the solar system, forming a resource-rich, oxygenated habitable Earth. Our civilization is predicated on stable climate and coastlines, yet the geological record reveals numerous episodes of enormous change, innovation, radiation and extinction. We need to understand the underlying mechanisms and feedbacks to predict possible future paths of the Earth. How do the evolution of continents and climates affect ocean circulation, shifts in coastlines, rainfall, rock weathering and soil formation or erosion? How long may it take for geological forces, as opposed to humans, to dominate Earth's surface evolution again in the future? Human civilisation equally depends on a range of metal resources to build renewable energy infrastructure. To identify portions of the Earth's crust rich in these resources, we need to understand how bursts in mineralisation are related to the interplay between the evolution of the deep Earth, the crust and surface environments. Recent advances in computing power, software development and global database development enable rapid advances in modelling the interconnectivity of Earth system processes, transforming our ability to understand past and possible future evolutionary paths of our planet.

Prof. Dietmar Muller is the founder of **EarthByte** at the School of Geosciences, University of Sydney. He is a world-leading geophysicist whose research has transformed our understanding of the Earth's evolution over the past 200 million years. Prof. Muller has developed new capabilities for modelling tectonic, geodynamic, environmental and resource-formation processes.

Geraint Lewis 1:45pm – 2:15pm

What's happening in the dark-side of the universe?

Abstract.

Geraint Lewis is a Professor of Astrophysics at the **Sydney Institute for Astronomy, University of Sydney**. He is head of the Gravitational Astrophysics Group, is the Associate Head for Research, and Deputy Director of the Sydney Informatics Hub. He is a Welsh astrophysicist, who is best known for his work on dark energy, gravitational lensing and galactic cannibalism.



Caroline Foster 2.15pm – 2.45pm

Solving the century old challenge of measuring the true shape of galaxies

The nature of astronomical distances is such that even our nearest neighboring galaxy is too far to explore by probe. Much like shadow puppetry, astronomers see galaxies in projection. Since distinct shapes project identically in 2D, measuring the true 3D (or intrinsic) shape of galaxies is a challenging problem that is nearly a century old. I will describe the problem and show how recent technological advances are finally allowing astronomers to reliably measure the intrinsic shape of galaxies.

Dr Caroline Foster is the **Astro3D Centre for Research Excellence** fellow at the University of Sydney. Her research focuses on galaxy formation and evolution through the analysis of the 3D structure, dynamics and chemistry of galaxies.



Cormac Purcell 2:45pm - 3:15pm

**Sharks, tombs and cosmic bubbles:
how machine-learning spans diverse
scientific fields**



Abstract

Cormac Purcell a researcher and lecturer in the **Department of Physics and Astronomy** at Macquarie University. He uses radio telescopes to investigate the interstellar medium, with a focus on magnetism and star-formation in the Milky Way.

Humans and Society 3:30pm-5:30pm

Kai Riemer and Sandra Peter 3.30pm – 3.55pm

AI - History, hype and hysteria

Abstract

Kai Riemer is the Professor of Information Technology and Organisation, Business Information Systems, Sydney Business School. He is the founder and leader of the Digital Disruption Research Group (DDRG) and convener of DISRUPT.SYDNEY™. His expertise spans the fields of digital commerce, inter-firm networking, the future of work, technology adoption, and the philosophy of technology.



Sandra Peter is the Director for Sydney Business Insights with expertise in disruptive innovations for education and the future of business.

Simon Hammond 3.55pm – 4.20pm

Understanding speech and text in low-resource languages

The Low Resource Languages for Emergent Incidents (LORELEI) program aims to automatically detect emergencies and needs for international aid in languages from Haitian Creole to Tamil. But language technology has come to rely on huge amounts of training data for its accuracy; how can we adapt these engines to languages with few existing resources?

Bing Ong 4.20pm – 4.45pm

Democratising AI for businesses

AI start-up, Daisee, saw a gap in the Australian market, with few specialist AI companies developing software for businesses of any size to use. It's focus is on developing software while helping companies quickly integrate customised AI capabilities. Daisee's data scientist, Bing Ong, will talk about the road to developing its first product, and how opening up AI for businesses is going to change the world in which we live, starting from the call center.

Bing is currently a data scientist at Daisee, a local start-up that specialises in artificial intelligence. Bing has over seventeen years of experience in IT, working for startups to MNCs (eg. Oracle (nearly ten years)) with experience spanning across anything to do with data – presales, system integration, project management and program development. She found her passion in data science after recently completing her studies in the Masters of Data Science and Innovation with UTS. In her new role at Daisee, she is continuing her knowledge-seeking journey by working with world-class consultants and talented data scientist to provide value to clients.



Sonya Corcoran 4:45pm – 5:10pm

Digital Innovation - Who is making what and how?

Abstract

Sonya Corcoran is the manager for Digital Innovation. She leads the **TechLab** where students and researchers can experiment with new technology and algorithms. She is also a co-founder of **Changineers** where she helps to design experiential learning technology.



Roman Marchant 5:10pm - 5:35pm

**Understanding and preventing crime
using Data Science**

Abstract

Dr Roman Marchant is a Research Fellow at the Centre for Translational Data Science. He applies data science to predict crime and to understand criminal behaviour. His work informs governmental policies and policing programs which are designed to curb domestic violence levels within the community.

